EDC · MOCK

EDC mock B

Practice questions · Day 28 · Fri 20 Nov · pack EX2

Instructions

- Time allowed: **2 hours**, strictly timed. Sit it in one go, the day before the exam.
- Open book (hard copy notes, your reference sheet, textbooks). Calculator allowed. No internet.
- **11 questions, 80 marks. Answer all of them.** The marks for each part are in square brackets.
- Show your working. Write the formula first, then the numbers, then the answer with units. Most marks are for method.
- The paper is about 45 % diodes and BJTs (Q1 part, Q2–Q6) and about 55 % FETs, multistage, op-amp and power (Q1 part, Q7–Q11).
- Budget about **1.5 minutes per mark**. If a part has taken twice its budget, write down what you would do next and move on.
- Unless a question says otherwise: Si diodes and BJTs have $V_D = V_{BE} = 0.7\ \text{V}$, $V_{CE,\text{sat}} = 0.2\ \text{V}$, $r_e = 26\ \text{mV}/I_E$, capacitors are short circuits at signal frequency, and r_o is ignored for BJTs.
- Every circuit is described in words. Draw it yourself before you start each question. The drawing is part of the method.

Question	Topic	Marks	Time budget
1	Multiple choice (8 parts)	8	10 min
2	Semiconductor physics; diode r_d and temperature	7	10 min
3	Two-level clipper and biased clamper	5	8 min
4	Bridge rectifier with capacitor filter (design)	5	8 min
5	pnp voltage-divider bias, β sensitivity, saturation	8	12 min
6	Collector-feedback CE and common-base amplifiers	8	12 min
7	BJT current mirror	5	7 min
8	D-MOSFET with positive V_{GS}	8	12 min
9	JFET source follower and E-MOSFET drain-feedback amplifier	12	18 min
10	Two-stage JFET cascade	7	11 min
11	Power amplifiers: transformer-coupled class A and class B	7	10 min
	Total	**80**	**118 min**

Formula table

Semiconductors and diodes

Resistance, conductivity	$R = \dfrac{\rho L}{A}$, $\rho = \dfrac{1}{\sigma}$, $\sigma = q(n\mu_n + p\mu_p)$
Mass action	$n_0 p_0 = n_i^2$
Intrinsic concentration	$n_i = BT^{3/2} e^{-E_g/(2kT)}$; Si: $B = 7.3 \times 10^{15}\ \mathrm{cm^{-3}K^{-3/2}}$, $E_g = 1.12$ eV, $k = 8.62 \times 10^{-5}$ eV/K
Thermal voltage	$V_T = \dfrac{kT_K}{q}$, $k = 1.38 \times 10^{-23}$ J/K, $q = 1.6 \times 10^{-19}$ C
Shockley	$I_D = I_S(e^{V_D/nV_T} - 1)$
Diode small-signal resistance	$r_d = \dfrac{nV_T}{I_D}$
Si temperature rules	V_D falls about 2.5 mV/°C at fixed current; I_S doubles every 10 °C
Full-wave ripple (p-p)	$V_r = \dfrac{V_P}{2fCR} = \dfrac{I_L}{2fC}$, $I_L \approx \dfrac{V_P}{R}$, $V_{dc} = V_P - \frac{1}{2}V_r$
Conduction angle	$\omega \Delta t \approx \sqrt{2V_r/V_P}$
Full-wave diode currents	$i_{Dav} = I_L\left(1 + \pi\sqrt{\dfrac{V_P}{2V_r}}\right)$, $i_{Dmax} = I_L\left(1 + 2\pi\sqrt{\dfrac{V_P}{2V_r}}\right)$
Bridge PIV	$\mathrm{PIV} = V_{s,\mathrm{peak}} - V_D$

BJT

Currents	$I_C = \beta I_B$, $I_E = (\beta+1) I_B$, $\alpha = \dfrac{\beta}{\beta+1}$
Voltage-divider bias (exact)	$I_B = \dfrac{V_{Th} - V_{BE}}{R_{Th} + (\beta+1)R_E}$ (npn); pnp: $I_B = \dfrac{V_{CC} - V_{EB} - V_{Th}}{R_{Th} + (\beta+1)R_E}$
Collector-feedback bias	$I_B = \dfrac{V_{CC} - V_{BE}}{R_F + \beta R_C}$ (no R_E)
Collector feedback (AC)	$Z_i = \dfrac{r_e}{\frac{1}{\beta} + \frac{R_C}{R_C + R_F}}$, $Z_o \approx R_C \parallel R_F$, $A_v = -\dfrac{R_C \parallel R_F}{r_e}$
Common base	$Z_i = R_E \parallel r_e$, $Z_o = R_C$, $A_v = \dfrac{\alpha (R_C \parallel R_L)}{r_e}$
Source loading	$A_{v_s} = \dfrac{Z_i}{Z_i + R_s} A_{v_L}$
Current mirror	$I_o = \dfrac{I_{REF}}{1 + 2/\beta}$

FETs

JFET and D-MOSFET	$I_D = I_{DSS}\left(1 - \dfrac{V_{GS}}{V_P}\right)^2$
E-MOSFET	$I_D = k(V_{GS} - V_{GS(Th)})^2$, $g_m = 2k(V_{GSQ} - V_{GS(Th)})$
Transconductance	$g_{m0} = \dfrac{2I_{DSS}}{\lvert V_P \rvert}$, $g_m = g_{m0}\left(1 - \dfrac{V_{GSQ}}{V_P}\right) = g_{m0}\sqrt{\dfrac{I_{DQ}}{I_{DSS}}}$, $r_d = \dfrac{1}{y_{os}} = \dfrac{1}{g_{os}}$
CS (bypassed R_S)	$Z_i = R_G$ or $R_1 \parallel R_2$, $Z_o = r_d \parallel R_D$, $A_v = -g_m(r_d \parallel R_D)$
Source follower	$Z_i = R_G$, $Z_o = r_d \parallel R_S \parallel \dfrac{1}{g_m}$, $A_v = \dfrac{g_m(r_d \parallel R_S)}{1 + g_m(r_d \parallel R_S)}$
Drain feedback	$Z_i = \dfrac{R_F + r_d \parallel R_D}{1 + g_m(r_d \parallel R_D)}$, $Z_o = R_F \parallel r_d \parallel R_D$, $A_v = -g_m(R_F \parallel r_d \parallel R_D)$
Cascade	$A_{vT} = A_{v1}A_{v2}$, each stage loaded by the next stage's Z_i

Op-amp, digital and power

CMRR	$\mathrm{CMRR} = \dfrac{A_d}{A_c}$, $\mathrm{CMRR(dB)} = 20\log_{10}\dfrac{A_d}{A_c}$
Noise margins	$NM_L = V_{IL} - V_{OL}$, $NM_H = V_{OH} - V_{IH}$
Transformer	$a = \dfrac{N_1}{N_2}$, $R'_L = a^2 R_L$
Class A (transformer-coupled)	$P_i = V_{CC} I_{CQ}$; maximum ideal $P_o = \frac{1}{2} V_{CC} I_{CQ}$, $\eta_{max} = 50\%$
Class B push-pull	$P_i = \dfrac{2V_{CC}V_{L(p)}}{\pi R_L}$, $P_o = \dfrac{V_{L(p)}^2}{2R_L}$, $P_{2Q} = P_i - P_o$, $\eta_{max} = 78.5\%$

Question 1 [8 marks]

Circle the correct answer in each part (I to VIII). [1 mark each]

I. Silicon at 300 K ($n_i = 1.5 \times 10^{10}$ cm^{-3}) is doped with donors at $N_D = 10^{16}$ cm^{-3}. The hole concentration is about:

a. 1.5×10^{10} cm^{-3} b) 2.25×10^{4} cm^{-3} c) 10^{16} cm^{-3} d) zero

II. A bridge rectifier and a centre-tapped full-wave rectifier are each designed to give the same peak output V_m. Compared with the centre-tapped circuit, each diode in the bridge must withstand a peak inverse voltage of about:

a. twice as much b) the same c) half as much d) zero, because the bridge diodes never reverse-bias

III. A BJT is in saturation. Which statement is true?

a. Only the base–emitter junction is forward biased, and $I_C = \beta I_B$
b. Both junctions are forward biased, and $I_C < \beta I_B$
c. Both junctions are reverse biased, and $I_C \approx 0$
d. Only the collector–base junction is forward biased, and $I_C > \beta I_B$

IV. An n-channel depletion-type MOSFET with $I_{DSS} = 6$ mA and $V_P = -3$ V is biased at $V_{GS} = +1$ V, with V_{DS} large. Then:

a. the device is cut off
b. $I_D < I_{DSS}$ because the gate is positive

c. the device is in enhancement mode and $I_D > I_{DSS}$
d. the gate draws a large current because the gate junction is forward biased

V. A CMOS inverter running from 5 V has $V_{OH} = 4.8$ V, $V_{OL} = 0.2$ V, $V_{IL} = 2.1$ V and $V_{IH} = 2.9$ V. Its noise margins are:

a. $NM_L = 2.1$ V, $NM_H = 2.9$ V b) $NM_L = NM_H = 1.9$ V c) $NM_L = NM_H = 0.8$ V d) $NM_L = 4.6$ V, $NM_H = 0.8$ V

VI. A differential amplifier has differential gain $A_d = 4000$ and common-mode gain $A_c = 0.4$. Its CMRR is:

a. 40 dB b) 60 dB c) 80 dB d) 10 000 dB

VII. The main cause of crossover distortion in a class B push-pull output stage is that:

a. the output transformer saturates at high current
b. neither transistor conducts until its base–emitter voltage reaches about 0.7 V, so there is a dead band near zero
c. both transistors conduct for the full cycle
d. the supply voltage is too high

VIII. A basic two-transistor BJT current mirror uses matched transistors with $\beta = 50$. The ratio I_o/I_{REF} is about:

a. 0.50 b) 0.962 c) 0.980 d) 1.04

Question 2 [7 marks]

(a) A silicon bar is 2 mm long with a square cross-section of 0.1 mm × 0.1 mm. It is doped with boron at $N_A = 5 \times 10^{15}$ cm^{-3}. Use $\mu_n = 1350$ cm²/Vs, $\mu_p = 500$ cm²/Vs and, at 300 K, $n_i = 1.5 \times 10^{10}$ cm^{-3}.

1. Find the hole and electron concentrations at 300 K. [1]
2. Find the conductivity and the resistance of the bar between its two end faces at 300 K. [2]
3. Use the n_i formula in the table to find n_i at 200 °C. Is the bar still extrinsic (that is, is $p \approx N_A$ still a good assumption)? Justify with a number. [1]

Question 2(b) circuit: 5 V supply and 20 mV signal in series with 4.3 kOhm and a diode

(b) A silicon diode with $n = 1$ is connected in series with a resistor $R = 4.3$ kΩ across a DC supply $V_{DD} = 5$ V (anode towards the supply, cathode to ground). A small sinusoidal signal of 20 mV peak is connected in series with the 5 V supply. Use $V_D = 0.7$ V for the DC analysis at 27 °C.

1. At 27 °C, find the DC diode current, the small-signal resistance r_d, and the peak ac voltage across the diode. [2]
2. The temperature rises to 77 °C. Using the course temperature rule for V_D, find the new DC current, the new r_d and the new peak ac voltage across the diode. Explain in one sentence why r_d went up even though I_D went up. [1]

Question 3 [5 marks]

Use the constant-voltage diode model with $V_D = 0.7$ V.

Question 3(a) circuit: two-level parallel clipper with 4 V and 6 V batteries

(a) Two-level parallel clipper. The input $v_i = 12 \sin \omega t$ V is applied to one end of a 1 kΩ series resistor. The other end of the resistor is the output node v_o, which has no load. Two branches go from the output node to ground:

- Branch 1: diode D1 with its **anode at the output node** and its cathode at the **positive** terminal of a 4 V battery. The battery's negative terminal is grounded.
- Branch 2: diode D2 with its **cathode at the output node** and its anode at the **negative** terminal of a 6 V battery. The battery's positive terminal is grounded.

Find the two clipping levels and the phase angles (ωt in degrees) at which each clipping starts and stops. Describe the output waveform over one cycle in words, as you would label a sketch. [3]

Question 3(b) circuit: biased clamper with a 2 V battery

(b) Biased clamper. A ±8 V, 1 kHz square wave is applied to a 1 µF capacitor in series. The other plate of the capacitor is the output node. From the output node to ground there is a 100 kΩ resistor, and also a diode with its **anode at the output node** and its cathode at the **positive** terminal of a 2 V battery, whose negative terminal is grounded. Find the two output levels and the peak-to-peak output. Describe the output waveform in words. State why the 100 kΩ resistor does not upset the result. [2]

Question 4 [5 marks]

Question 4 circuit: bridge rectifier with capacitor filter and 150 ohm load

A bridge rectifier (four Si diodes, 0.7 V each) is fed from a transformer secondary of 12 V rms at 50 Hz. The rectifier output drives a filter capacitor C in parallel with a 150 Ω load.

1. Find the peak output voltage V_P. [1]
2. Find the minimum C that keeps the peak-to-peak ripple at or below 0.8 V. [1]
3. You fit the next standard value, 1500 µF. Find the actual ripple and the DC output voltage. [1]
4. Find the peak diode current i_{Dmax} (use $I_L \approx V_P/R$). [1]
5. Find the peak inverse voltage each diode must withstand. [1]

Question 5 [8 marks]

Question 5 circuit: pnp voltage-divider bias from +12 V

A **pnp** BJT is biased with a voltage divider from a single +12 V supply. The emitter connects through $R_E = 1.5$ kΩ up to the +12 V rail. The collector connects through $R_C = 2.7$ kΩ down to ground. $R_1 = 15$ kΩ runs from the +12 V rail to the base, and $R_2 = 47$ kΩ runs from the base to ground. Take $V_{EB} = 0.7$ V, $V_{EC,\text{sat}} = 0.2$ V and $\beta = 100$.

(a) Find the Thevenin equivalent seen from the base, then find I_C, V_C and V_{EC} using the exact (Thevenin) method. Confirm the transistor is active. [4]

(b) The transistor is replaced by others with $\beta = 50$ and $\beta = 200$. Find I_C for each and the percentage change from the $\beta = 100$ value. Comment on the stability of this bias. [2]

(c) With $\beta = 100$, R_C is changed to 8.2 kΩ. Show that the active assumption now fails, and find the actual collector current. [2]

Question 6 [8 marks]

Question 6(a) circuit: collector-feedback CE amplifier

(a) Collector feedback. An npn BJT has its emitter grounded. $R_C = 2.2$ kΩ runs from $V_{CC} = 10$ V to the collector, and $R_F = 180$ kΩ runs from the collector to the base. The input signal is capacitively coupled to the base and the output is taken from the collector through a coupling capacitor. $\beta = 150$. Find I_E, r_e, Z_i, Z_o and A_v (no load). [4]

Question 6(b) circuit: common-base amplifier with R_E to -4 V

(b) Common base. An npn BJT has its base grounded. The emitter connects through $R_E = 1.2$ kΩ to a -4 V supply. The collector connects through $R_C = 3.3$ kΩ to +12 V. A signal source v_s with internal resistance $R_s = 50\ \Omega$ is capacitively coupled to the emitter. The collector is capacitively coupled to a load $R_L = 6.8$ kΩ. $\alpha = 0.99$. Find I_E, r_e, Z_i, the loaded gain $A_{v_L} = v_o/v_i$ and the overall gain $A_{v_s} = v_o/v_s$. State the phase of v_o relative to v_s. [4]

Question 7 [5 marks]

Question 7 circuit: two-transistor current mirror with emitters on -9 V

A current mirror uses two matched npn transistors, Q1 and Q2, with $\beta = 100$. Both emitters connect to a -9 V rail. The bases are tied together and Q1 is diode-connected (its collector is tied to its base). A resistor R runs from ground (0 V) to Q1's collector. Q2's collector draws the output current I_o from ground through a load resistor R_L.

1. Find R so that $I_o = 2.00$ mA. [2]
2. Find the largest R_L for which Q2 stays active. [2]
3. With R kept at your value from part 1, the transistors are replaced by a matched pair with $\beta = 50$. Find the new I_o. [1]

Question 8 [8 marks]

Question 8 circuit: D-MOSFET voltage-divider common-source amplifier

An n-channel **depletion-type MOSFET** has $I_{DSS} = 8$ mA, $V_P = -4$ V and $y_{os} = 10$ µS. It is used in a voltage-divider common-source amplifier with $V_{DD} = 18$ V. $R_1 = 100$ MΩ runs from V_{DD} to the gate and $R_2 = 20$ MΩ from the gate to ground. $R_D = 1.0$ kΩ runs from V_{DD} to the drain. $R_S = 300\ \Omega$ runs from the source to ground and is fully bypassed by a capacitor C_S. The input is capacitively coupled to the gate and the output is taken from the drain through a coupling capacitor.

(a) Find V_G, then solve for I_{DQ} and V_{GSQ}. Say which root you reject and why. [3]

(b) Find V_D, V_S and V_{DS}, and confirm the device is in saturation (the pinch-off region). [1]

(c) Find g_{m0} and g_m. Explain in one sentence why $g_m > g_{m0}$ here. [1]

(d) Find Z_i, Z_o and A_v. Then find the gain when a 3.3 kΩ load is capacitively coupled to the drain. [3]

Question 9 [12 marks]

Question 9(a) circuit: JFET source follower

(a) JFET source follower. An n-channel JFET has $I_{DSS} = 10$ mA, $V_P = -4$ V and $y_{os} = 20$ µS. The drain connects directly to $V_{DD} = 15$ V. The gate connects to ground through $R_G = 1$

MΩ, and the input is capacitively coupled to the gate. The source connects to ground through $R_S = 2$ kΩ, and the output is taken from the source through a coupling capacitor.

1. Find I_{DQ} and V_{GSQ}. [2]
2. Find g_m. [1]
3. Find Z_i and Z_o. [1]
4. Find A_v with no load, and with a 1 kΩ load coupled to the source. [1]
5. With the 1 kΩ load and $v_i = 0.5$ V peak, find the peak output voltage and the peak v_{gs}. [1]

Question 9(b) circuit: E-MOSFET drain-feedback amplifier

(b) E-MOSFET drain-feedback amplifier. An n-channel enhancement MOSFET has $k = 0.5$ mA/V², $V_{GS(Th)} = 2$ V and $r_d = 50$ kΩ. The source is grounded. $R_D = 2$ kΩ runs from $V_{DD} = 12$ V to the drain, and $R_F = 10$ MΩ runs from the drain to the gate. The input is capacitively coupled to the gate and the output is taken from the drain through a coupling capacitor.

1. Find V_{GSQ}, I_{DQ} and V_{DSQ}. Say why this device is guaranteed to be in saturation. [2]
2. Find g_m. [1]
3. Find Z_i, Z_o and A_v. [3]

Question 10 [7 marks]

Question 10 circuit: two identical self-bias JFET stages, RC coupled

A two-stage RC-coupled amplifier uses two identical n-channel JFETs with $I_{DSS} = 8$ mA, $V_P = -4.5$ V and $r_d = 40$ kΩ. Each stage is an identical self-bias common-source stage on $V_{DD} = 18$ V: $R_G = 2.2$ MΩ from gate to ground, $R_S = 1$ kΩ from source to ground (fully bypassed), and $R_D = 3.3$ kΩ from V_{DD} to drain. The drain of stage 1 is capacitively coupled to the gate of stage 2. The drain of stage 2 is capacitively coupled to a load $R_L = 10$ kΩ. The input is a source of 20 mV rms with internal resistance $R_{sig} = 10$ kΩ, coupled to the gate of stage 1.

1. Find I_{DQ}, V_{GSQ} and V_{DSQ} (the same for both stages). [2]
2. Find g_m. [1]
3. Find the loaded gain of each stage, A_{v1} and A_{v2}. [2]
4. Find the overall gain v_o/v_{sig} and the rms output voltage, plus the amplifier's Z_i and Z_o. [2]

Question 11 [7 marks]

Question 11(a) circuit: transformer-coupled class A into two 8 ohm speakers in parallel

(a) A transformer-coupled class A amplifier runs from $V_{CC} = 12$ V with a quiescent collector current $I_{CQ} = 150$ mA. The load is two 8 Ω loudspeakers in parallel on the transformer secondary. For the largest possible symmetric collector swing, the reflected load must be $R'_L = V_{CC}/I_{CQ}$.

1. Find the required turns ratio $N_1 : N_2$. [2]
2. Find the maximum ideal output power and the efficiency at that point. [1]

Question 11(b) circuit: class B push-pull, V_CC = 20 V, 8 ohm load

(b) A class B push-pull amplifier runs from $V_{CC} = 20$ V (each transistor sees the full 20 V) and drives an 8 Ω load directly.

1. Find the maximum output power and the efficiency at that output. [2]
2. Find the maximum total transistor dissipation, the dissipation in each transistor, and the peak load voltage $V_{L(p)}$ at which it occurs. What is the efficiency at that point? [2]

End of paper.